Global Facilities - Design & Construction Standard - Mechanical - Compressed Air System (CDA, HPCDA, XCDA, ICA)

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1.0 Purpose

1.1 This document serves as a System Standard document that provides specifications that shall be followed during the Design, Engineering and Construction of new Compressed Air Systems as part of new Greenfield FAB Construction projects

1.2 This Standard is applicable for Clean Dry Air (CDA), High Pressure Clean Dry Air (HPCDA), Purified Clean Dry Air (XCDA) and Instrument Control Air (ICA) systems.

1.3 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.

2.0 Scope

2.1 The document covers the new Basebuild System installations only and does NOT cover tool installation.

2.2 Site(s) impacted

2.2.1 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects.

3.0 Definitions and Acronyms

- **AMC -** Airborne Molecular Contaminants
- **CDA** – Clean Dry Air
- **CHW -** Chilled Water
- **CW -** Condenser Water
- **CCW -** Compressor Cooling Water
- **CD -** Condensate Drain
- **FAT –** Factory Acceptance Test
- **FMCS/SCADA -** Facility Monitoring and Control System / Supervisory Control and Data Acquisition
- **FCT** – Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
- **GFTT** – Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
- **GFC** – Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
- **Global Facilities Central Team** – This is used interchangeably with Innovation & Integration Team
- **Global Facilities Construction Team** – This is used interchangeably with Global Construction Team
- **Global Facilities Operations Support Team** – This is used interchangeably with Global Facilities Operations Team
- **Global Facilities Planning Team** – This is used interchangeably with Global Facilities Line Planning Team
- **HPCDA (PCDA) -** High Pressure Clean Dry Air (Photo Clean Dry Air)
- **ICA -** Instrument Control Air
- **ICW -** Industrial City Water
- **IST –** Integrated System Test
- **ITP –** Inspection and Test Plan
- **LOTO/COHE -** Lockout Tagout / Control of Hazardous Energy
- **MAU -** Makeup Air Unit
- **N+1+F -** System sizing concept for equipment quantity consisting of N duty units, +1 redundant standby unit, +1 future unit
- **OAT –** Operational Acceptance Test
- **PDP -** Pressure Dew Point
- **PLC -** Programmable Logic Controller
- **POC/LPOC -** Point of Connection / Lateral Point of Connection
- **SAT –** Site Acceptance Test
- **TUM -** Tool Utility Matrix
- **UPS -** Uninterruptible Power Supply
- **VSD/VFD -** Variable Speed Drive / Variable Frequency Drive
- **XCDA -** Purified Clean Dry Air

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4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT Lead (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

5.0 Overview of Systems

5.1 The following Compressed Air systems are used to support the requirements from Manufacturing and Facilities loads:

5.1.1 **Clean Dry Air (CDA)** is compressed air used for driving pneumatic components and controls on manufacturing tools and support equipment.

5.1.2 **High Pressure CDA (HPCDA)** is compressed air that is supplied at a higher supply pressure and is used for driving pneumatic components and controls on Photolithography tools and equipment.

5.1.3 **Purified CDA (XCDA)** is compressed air (typically using HPCDA) that goes through a purifier to remove Acids, Amines, Organics, and other Airborne Molecular Contaminants (AMCs), and then is supplied to Photolithography tools.

5.1.4 **Instrument Control Air (ICA)** is compressed air that is supplied at a lower pressure than CDA, for driving control valves, dampers, instrumentation, and other components on Facilities systems.

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5.2 The main CDA and HPCDA equipment shall be as follows:
- 5.2.1 Compressors
- 5.2.2 Receiver Tanks
- 5.2.3 Aftercoolers
- 5.2.4 Dryers
- 5.2.5 Filters

5.3 The auxiliary CDA and HPCDA equipment and components shall be as follows:
- 5.3.1 Makeup Air unit for Air Intake of the Compressors
- 5.3.2 Exhaust Fans for Heat Exhaust discharged from the Compressors and Dryers
- 5.3.3 Compressor Cooling Water Pumps
- 5.3.4 Compressor Cooling Water Heat Exchangers

5.4 The equipment and components for XCDA and ICA shall be as follows:
- 5.4.1 XCDA Purifier Skids
- 5.4.2 N2 backup provisions for ICA

5.5 **Appendix 2** is the recommended provision of Compressed Air for pneumatic equipment of various Utility Systems.

6.0 Performance Spec Requirements:

6.1 The Compressed Air Systems shall be designed to achieve the operational performance specs as outlined in **Table 1** to **Table 5**:

Table 1: Operational Performance Specs for CDA

Parameter	Specification Requirement	Remarks
Supply Pressure	0.80 ± 0.028 MPa (116 ± 4 psig)	Measured at Supply Lateral End of Line (at Subfab Level)
Pressure Dew Point	< -40 °C PDP (-40 °F PDP)	Measured at Main Supply Header After the Dryers

Table 2: Operational Performance Specs for HPCDA:

Parameter	Specification Requirement	Remarks
Supply Pressure	0.95 ± 0.028 MPa (138 ± 4 psig)	Measured at Supply Lateral End of Line (at Subfab Level)
Pressure Dew Point	< - 40 °C PDP (-40 °F PDP)	Measured at Main Supply Header After the Dryers

Table 3: Operational Performance Specs for XCDA:

Parameter	Specification Requirement	Remarks
Supply Pressure	0.876 ± 0.03 MPa (127 ± 4.3 psig)	Measured at Supply Lateral End of Line (at Subfab Level)
Moisture	< 100 ppt H2O	Measured at XCDA Purification Unit
Acids	< 1 ppt	Measured at XCDA Purification Unit
Amines	< 10 ppt	Measured at XCDA Purification Unit
Condensables	< 1 ppt	Measured at XCDA Purification Unit
Refractories	< 1 ppt	Measured at XCDA Purification Unit

Table 4: Operational Performance Specs for ICA:

Parameter	Specification Requirement	Remarks
Supply Pressure	> 0.65 MPa (94.3 psig)	Measured at Main Supply Header (after Pressure Regulating Valve)

Table 5: CDA, HPCDA, and XCDA Quality Certification

Parameter	Specification Requirement	Remarks
Particles	Exceeds ISO 8573-1:2010 Class 1 < 353 counts/m3 @ 0.1 to 0.5µm (< 10 counts/ft3 @ 0.1 to 0.5µm)	
Moisture Content	Exceeds ISO 8573-1:2010 Class 2 < -40 °C (°F) Dewpoint	-40 °C Dewpoint equivalent to 14.5 ppm H2O @ 0.80 MPa / 116 psig
Oil Content	ISO 8573-1:2010 Class 1 < 0.01 mg/m3	

7.0 Equipment Design and Sizing Requirements:

7.1 **Redundancy and Future** – CDA and HPCDA system main equipment shall be designed for N+1+F; for Compressors, Aftercoolers, Dryers, Filters, and Compressor Cooling Water Pumps. All main system equipment shall have N+1 redundancy (with one redundant unit having capability as online standby), and a provision to install one additional future unit (F) for future capacity expansion.

7.2 **System Diversity and Sizing Factor** – CDA and HPCDA systems shall be sized so that the N capacity meets 115-120% of the CDA demand at full build (after initial construction of the system), based on the manufacturing tool layout and TUM. Therefore, the CDA and HPCDA system sizing for N capacity shall have 15-20% spare future capacity over the CDA and HPCDA demand at full build (after initial construction of the system).

7.3 To determine the CDA and HPCDA demand for N at full build, an approximate diversity factor of 70 - 80% of the Peak demand from the TUM. The diversity factor can be adjusted via each site's engineering judgement based on previous known diversity factors, or by interpolating a value between the Average and Peak demand from the TUM. In addition, the CDA and HPCDA systems shall be sized to include the capacity of one additional future unit (F). See **Table 6.**

Table 6: CDA and HPCDA System Sizing

- CDA and HPCDA System Demand at Full Build,
  D = TUM Peak Demand x 70-80% Diversity
- CDA and HPCDA System N capacity,
  N = 115-120% x D
- CDA and HPCDA System Sizing = N + F

7.4 Equipment Quantity – CDA System main equipment shall be selected for a minimum of 3 units (2+1) or more, with each unit to be equal in capacity sizing; for Aftercoolers, Dryers, and Filters. CDA compressors shall be selected as follows:

7.4.1 Base Load CDA Compressors shall be Centrifugal type Compressors.

7.4.2 Two (2) CDA Compressors shall be Trim Load Compressors, sized at 50% capacity each of one Base Load Compressor. The Trim Load Compressors can be Centrifugal type or Rotary Screw type (either Fixed Speed or with Variable Speed Drive).

7.5 For CDA, the larger Centrifugal type Compressors shall be used as the units for base load operation, and the smaller 50% Rotary Screw or Centrifugal Compressors shall be used as trim units to meet changes in peak demand loading, and during system startup/initial loading.

7.6 HPCDA System equipment can be selected for a minimum of 2 units (1+1), or more, with each unit to be equal in capacity sizing. The HPCDA demand at full build shall be split evenly across the number of equally sized equipment.

7.6.1 HPCDA Compressors can be Centrifugal type or Rotary Screw type Compressors (either Fixed Speed or with Variable Speed Drive).

7.6.2 Because HPCDA systems normally have relatively small load and low consumption, 2 Compressor units (1+1) is typically sufficient for the loading. If the load is much higher, than it is recommended to use higher number of units (minimum 3 units instead).

7.7 **Equipment Arrangement** – All Main CDA and HPCDA equipment shall be arranged in parallel together and connected via a common piping header for each equipment set; for Compressors, Aftercoolers, Receiver Tanks, Dryers, and Filters. This allows any combination of equipment to be in operation concurrently.

7.8 **Equipment Layout** – Equipment shall be suitably located to comply with all equipment maintenance clearances and access requirements. Adequate space and a clear routing path shall be reserved for future move-in of new equipment and to allow move-in and move-out of large components during maintenance and repair (such as motors, compressor elements, filters etc.).

7.9 Equipment Selection Criteria - See Table 7.

Table 7: Equipment Selection Criteria

Equipment	Type and Description
<u>CDA</u> Rotary Screw Compressor, Oil Free, Water-Cooled, 2 Stage Elements, Fixed Speed or with Variable Speed	• Trim Load Compressors can be Rotary Screw Type, Oil-Free, Water-Cooled, Fixed Speed or with Variable Speed Drive, with 2 stage elements: • Each Trim Load Rotary Screw Compressor shall be sized at 50% capacity of the larger Centrifugal Compressor size. • Compressor shall consist of the following components:

Equipment	Type and Description
Drive	○ Air intake section with air filters ○ Main drive motor and drive coupling ○ 2-stage compressor elements ○ 2-stage intercoolers ○ Oil pump and circulation system ○ Variable Speed Drive (Optional) ○ Discharge Check Valve ○ Flexible connection/bellows on Discharge ○ Drain Assembly ○ Compressor-mounted Local Control Panel with display • Compressor sizing and selection shall be as follows: ○ Compressor shall be selected with a minimum discharge pressure capability of 0.96 MPa (139.2 psig) **Note -** actual operation of CDA Supply Pressure from Compressors may be lower than equipment rating of 0.96 MPa, to achieve the required CDA system pressure. ○ Compressor shall have minimum capacity turndown of 50% of the design flowrate (so that compressor supply flowrate can be reduced to 50% or lower of the compressor's rated flow). ○ Compressor motor size shall be selected to satisfy the loading per all operating points on the performance curve • Compressor shall be specified with the following options/accessories: - ○ Sound attenuation enclosure ○ Motor efficiency rating shall be minimum IE3 ○ Motor IP rating shall be minimum IP55 • Compressor technical submittal shall include the following information: ○ Performance curve indicating the design operation point (flow, pressure, and power consumption, based on part loading or part VSD speed) ○ Performance curve indicating the maximum operating point (flow, pressure, and power consumption, based on full loading or full VSD speed)

Equipment	Type and Description
	○ Performance curve at different operating speeds 25%, 50%, 75% (flow, pressure, and power consumption, based on part loading or part VSD speeds) ○ Compressor Cooling Water requirements ○ Compressor Turndown rate ○ Compressor Sound Level Performance • Compressor installation shall be as follows: ○ Compressor CDA discharge piping shall be installed with full size isolation valves, pressure gauge, and temperature gauge ○ Condenser Water piping shall have a strainer on the inlet side, flow balancing valve on discharge side, and full size isolation valves, flexible piping, low point drain ports, temperature gauges, and pressure gauges installed on both inlet and discharge sides ○ Compressors shall be installed on minimum 4-inch (100mm) high concrete housekeeping pad ○ Compressor drains shall be equipped with zero-loss auto drain valves • Provide a separate electrical voltage transformer for the compressors, if required due to the compressor voltage rating. • Compressor sound level performance shall be < 80 dBA measured at a distance of 1.0m away from the compressor enclosure. **Note**: Approved Manufacturers – Ingersoll Rand, Atlas Copco, or approved equivalent
HPCDA Rotary Screw Compressor, Oil-Free, Water-Cooled, 2 Stage Elements, Fixed Speed or with Variable Speed Drive	• For HPCDA Compressors Rotary Screw Type, Oil-Free, Water-Cooled, Fixed Speed or with Variable Speed Drive, with 2 stage elements: • Compressor sizing and selection shall be as follows: ○ Compressor shall be selected with a minimum discharge pressure capability of 1.06 MPa (153.75 psig) **Note -** actual operation of HPCDA Supply Pressure from Compressors may be lower than equipment rating of 0.96

Equipment	Type and Description
	MPa, to achieve the required HPCDA system pressure. ○ Compressor shall have minimum capacity turndown of 50% (so that compressor supply flowrate can be reduced to 50% or lower of the compressor's rated flow). **Note**: HPCDA Rotary Screw Compressors shall be selected and installed with the same features as the CDA Rotary Screw Compressor. **Note**: Approved Manufacturers – Ingersoll Rand, Atlas Copco, or approved equivalent
CDA Centrifugal Compressor, Oil-Free, Water-Cooled, with 3-stage Elements	• The Base Load CDA Compressors shall be Centrifugal Type, Oil-Free, water-cooled, with 3 stage elements: • Trim Load CDA Compressors can also be Centrifugal Type. Each Trim Load Centrifugal Compressor shall be sized at 50% capacity of the larger Centrifugal Compressor size. • Compressors shall consist of the following components: ○ Air intake section with air filters ○ Main drive motor and drive coupling ○ 3-stage compressor elements with rotors, diffusers, and gear drive ○ 2-stage intercoolers and 1-stage aftercooler ○ Oil pump and circulation system ○ Blow-off valve with actuator ○ Inlet guide vanes or inlet butterfly valve with actuator ○ Silencer on vent line ○ Discharge Check Valve ○ Flexible connection/bellows on Discharge ○ Drain assembly ○ Compressor-mounted Local Control Panel with display • Compressor sizing and selection shall be as follows:

Equipment	Type and Description
	◦ Compressor shall be selected with a minimum discharge pressure capability of 0.96 MPa (139.2 psig) **Note -** actual operation of CDA Supply Pressure from Compressors may be lower than equipment rating of 0.96 MPa, to achieve the required CDA system pressure. ◦ Compressor shall have minimum capacity turndown of 25% of the design flowrate (so that compressor supply flowrate can be reduced to 75% or lower of the compressor’s design flowrate), without surge ◦ Compressor motor size shall be selected to satisfy the loading per all operating points on the performance curve • Compressor shall be specified with the following options/accessories: ◦ Sound attenuation enclosure ◦ Motor efficiency rating shall be minimum IE3 ◦ Motor IP rating shall be minimum IP55 • Compressor technical submittal shall include the following information: ◦ Performance curve indicating the design operation point (flow, pressure, and power consumption) ◦ Performance curve at different Inlet Guide Vane positions and operating points: (flow, pressure, and power consumption, based on different Inlet Guide Vane positions) ◦ Compressor Cooling Water requirements ◦ Compressor Turndown rate ◦ Compressor Sound Level Performance • Compressor installation shall be as follows: ◦ Compressor CDA discharge piping shall be installed with full size isolation valves, pressure gauge, and temperature gauge ◦ Condenser Water piping shall have a strainer on the inlet side, flow balancing valve on discharge side, and full-size isolation valves, flexible piping, low point drain ports, temperature gauges, and

<table>
<tr><th>Equipment</th><th>Type and Description</th></tr>
<tr><td></td><td>pressure gauges installed on both inlet and discharge sides
○ Compressors shall be installed on minimum 4-inch (100mm) high concrete housekeeping pad
○ Compressor drains shall be equipped with zero-loss auto drain valves

• Compressor sound level performance shall be < 80 dBA measured at a distance of 1.0m away from the compressor enclosure.

<u>Note</u>: Approved Manufacturers – Ingersoll Rand, Atlas Copco, F.S. Elliott, or approved equivalent</td></tr>
<tr><td><u>HPCDA</u>
Centrifugal Compressor, Oil-Free, Water-Cooled, with 3-stage Elements</td><td>• For HPCDA Compressors Centrifugal Type, Oil-Free, water-cooled, with 3 stage elements:
• Compressor sizing and selection shall be as follows:
○ Compressor shall be selected with a minimum discharge pressure capability of 1.06 MPa (153.75 psig)
Note - actual operation of HPCDA Supply Pressure from Compressors may be lower than equipment rating of 0.96 MPa, to achieve the required HPCDA system pressure.
○ Compressor shall have minimum capacity turndown of 25% of the design flowrate (so that compressor supply flowrate can be reduced to 75% or lower of the compressor’s design flowrate), without surge
○ Compressor motor size shall be selected to satisfy the loading per all operating points on the performance curve

<u>Note</u>: HPCDA Centrifugal Compressors shall be selected and installed with the same features as the CDA Centrifugal Compressor.

• <u>Note</u>: Approved Manufacturers – Ingersoll Rand, Atlas Copco, or approved equivalent</td></tr>
</table>

Equipment	Type and Description

Aftercoolers	• Aftercooler shall be located after the Compressor discharge, and prior to the Air Receiving Tank(s).
	• 2 Aftercooler shall be shell and tube type heat exchanger type.
	• Aftercooler sizing and selection shall be as follows:
	  ○ Leaving Air Temperature of CDA and HPCDA leaving the Aftercoolers shall be designed for capability of ~25-30°C. **Note -** actual operation of Aftercooler may achieve lower CDA Leaving Air Temperatures, for increased moisture removal and more efficient CDA Dryer Operation.
	  ○ Medium Temperature Chilled Water (MTCHW) shall be used as the cooling source to the aftercooler. Approximate MTCHW supply temperature is 12-13°C, and approximate MTCHW return temperature is 18-19°C, for a temperature difference of ~7°C
	• Aftercooler shall be stainless steel or carbon steel housing material, with copper heat exchanger tubing
	• Aftercooler installation shall be as follows:
	  ○ Strainer shall be installed on the Chilled Water inlet side
	  ○ Flow balancing valve shall be installed on Chilled Water return side
	  ○ Temperature gauges and pressure gauges shall be installed on Chilled Water supply and return side.
	  ○ Each Aftercooler Chilled Water control valve shall have a full-line size bypass with isolation valve, around the control valve. This is to allow maintenance and repair to be performed on the control valve, while keeping the Heat Exchanger in manual (bypass) operation.
	  ○ Aftercoolers shall be installed on minimum 4-inch (100mm) high concrete housekeeping pad

<table>
<tr><th>Equipment</th><th>Type and Description</th></tr>
<tr><td>Dryers</td><td>

Dryer shall be dual tower, heated-regenerative type, with zero purge loss for regeneration.
Dryer shall consist of the following components:

Dual drying chambers
Four-way switching valve at inlet between the dual drying chambers
Dryer-mounted Local Control Panel with Display
Filters integral with dryer skid
Vacuum pump or blower fan with inlet filter, for regeneration

If cooling water is required for Dryers, MT Chilled Water shall be used for Dryer Cooling Water.
Dryer technical submittal shall include the following information:

Volume flowrate at Dryer inlet and outlet
Maximum inlet temperature
Time duration for Dryer tower switchover
Time duration for drying/regeneration cycle
Dryer desiccant type
Dryer pressure drop performance
Dryer sound level performance
Dryer cooling water requirements (if required)

Dryers shall be Epoxy coated Carbon steel housings.
Dryers shall be installed with air vent port, drain port, and differential pressure gauge.
Dryer housings shall be rated for pressure vessel ASME Section VIII Division 1.
Dryers shall be installed on minimum 4-inch (100mm0 high concrete housekeeping pads.
Dryer shall have “Fail Safe” feature upon power loss. Upon recovery of power, the dryer and controller shall resume at the same stage where the power was lost.
Dryer shall never be programmed to by-pass the proper cooling steps to prevent over-heating of CDA air stream.

<u>Note</u>: Approved Manufacturers – Pneumatic Products/SPX/Nikkiso, Parker, Atlas
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Equipment	Type and Description
	Copco, Ingersoll Rand, Quan Yang Machinery, UFS, or approved equivalent
Filters	• Filters shall come integral with Dryer skid. • Filters shall include the following types: Pre-Filter, Carbon-Filter, After-Filter, and Final-Filter. • Each Filter shall be installed with air vent port, drain port, and differential pressure gauge. • 4 Filter housings shall be rated for pressure vessel ASME Section VIII Division 1. • Filter's rated temperature shall sufficiently be higher than the normal operating air temperature after the Dryer outlet to ensure it is functional. <u>Pre-Filters (Coalescent Filters):</u> ○ Filter shall be located upstream of dryer ○ Filter Element shall be coalescing type, with particle filtration rated at 99.99% efficient at 1.0μm particle size, and 99.99% efficiency of removing liquid water and oil droplets. ○ Housing Material shall be Epoxy coated Carbon Steel, or Stainless Steel. ○ Filter shall be equipped with zero-loss auto drain. <u>Carbon-Filters:</u> ○ Filter shall be located downstream of dryer ○ Filter Element shall be activated carbon type, rated to remove oil content to 0.003 mg/m3 ○ Housing Material shall be Epoxy coated Carbon Steel or Stainless Steel. <u>After-Filters:</u> ○ Filter shall be located downstream of dryer ○ Filter Element shall be rated for 99.99% efficient at 0.1μm particle size ○ Housing Material shall be Epoxy coated Carbon Steel, or Stainless Steel.

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<u>Final Filters:</u>

Filter shall be located downstream of dryer
Filter Element shall be rated for 99.99% efficient at 0.003μm particle size for gases and at 0.01 - 003μm particle size for liquids
Housing Material shall be Stainless Steel.

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<tr><td>Receiver Tank(s)</td><td>

Receiver Tank shall be located after the Aftercoolers, and prior to the Dryers.
Receiver Tank shall be vertical mounted type, with standalone legs.
Receiver Tank sizing and selection shall be as follows:

Receiver Tank capacity shall be minimum of 1.5m^3 volume per 1,000 CMH of Total CDA and HPCDA System Capacity (for N+F Compressors, including Future Units)

$$\text{CDA Tank Capacity } (m^3) = \text{CDA System Capacity (N+F, CMH)} \times \frac{1.5\ m^3}{1{,}000\ \text{CMH}}$$

$$\text{HPCDA Tank Capacity } (m^3) = \text{HPCDA System Capacity (N+F, CMH)} \times \frac{1.5\ m^3}{1{,}000\ \text{CMH}}$$

Receiver Tank shall be Stainless Steel or Epoxy coated Carbon Steel.
Tank shall be rated for pressure vessel ASME Section VIII Division 1.
CDA and HPCDA piping shall connect to the Receiver Tank at a low elevation on the inlet, and at a high elevation on the outlet.
The building structure and floor load rating shall be capable of allowing on-site hydrostatic pressure testing of the Receiver Tank to be performed (via water fill of the tanks) – for minimum 1 tank at a time.
Receiver Tank shall be equipped with the following accessories:

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Equipment	Type and Description
	○ Dual (2) safety relief valves, sized for 100% of the flow each ○ Low point drains and zero-loss auto drain valves ○ Pressure gauge and pressure transmitters ○ Manhole to allow inspection access
Compressor Central Controller	• A Central Controller Module shall be installed and connected to all Compressors, to determine the operating sequence and loading/unloading of the Compressors

7.10 **Compressor Cooling Water Heat Exchangers (For Heat Recovery)** – As the operating Compressors can generate a significant amount of heat, a common set of Heat Exchangers may be incorporated for heat recovery purposes. Nonetheless, the system design must be evaluated thoroughly on a case-by-case basis.

7.10.1 The heat recovery / cooling water system shall be studied for its feasibility to justify for the additional installation of heat exchangers, pumps, piping & accessories, spaces, etc. Furthermore, the system operation shall be evaluated in detail, for example, to consider for the additional pumping to overcome the system pressure loss and the maintenance effort for the additional complexity of the cooling water system.

7.10.2 The heat recovery / cooling water system shall incorporate sufficient redundant heat exchangers (N+1, N≥2, with one redundant unit having capability as online standby), and with common piping headers which allow for the combination that any heat exchanger can serve any of the Compressors.

7.11 **Compressor Cooling Water Pumps** – A common set of Compressor Cooling Water Pumps shall serve both the CDA and HPCDA compressors. The pumps shall be designed for N+1+F; the pumps shall have N+1 redundancy (with one redundant unit having capability as online standby), and a provision to install one additional future unit (F) for future capacity expansion. There shall be a minimum of 3 units (2+1) or more, with each unit to be equal in capacity sizing.

7.11.1 The CCW pumps shall be arranged in parallel together and connected via a common piping header, to serve both the CDA and HPCDA Compressors. This allows any combination of CCW pumps to serve any of the CDA and HPCDA Compressors.

7.11.2 Cooling Tower Condenser Water shall be the water source of the CCW pumps to supply to the Compressor cooling coils. Heat exchanger is not required due to its additional pressure loss, additional pumps at the cooled water side and its approach temperature will limit the cooling capability of condenser water.

7.11.3 Cooling Tower Condenser Water is the preferred water source to Chilled Water for the cooling of Compressors. As the operating temperature of Compressor is typically much higher than the temperature of Condenser Water, it will make economic sense by using Condenser Water instead of Chilled Water which is much more expensive to generate (its unit cost per kW cooling is approximately 5 or 6 times higher).

7.11.4 The CCW pump suction inlet shall be completed with strainer / suction diffuser to provide rough filtration by retaining debris / sand / particulate prior to the connection to the common supply header. The strainers are critical to protect the downstream equipment from the potential clogging of the cooling coil. The strainer mesh size shall properly be selected according to the water quality, pressure drop, etc., and to the manufacturer’s recommendation. Some typical range of mesh size are as follows:

i. Pipe size ≤ 4”(DN100)– Mesh size 10~12, dia. ~1.5mm
ii. 4”(DN100)<Pipe size ≤ 10”(DN250)– Mesh size ~6, dia. ~3.0mm

Note: Finer mesh size (e.g. Mesh size >20, dia. <0.08mm) should be considered when the water has a high concentration of debris or particulate and may cause damage to the downstream equipment (e.g. construction stage, etc.). Fine mesh size strainers require more frequent cleaning to maintain the pressure drop within the range.

7.12 **Equipment Selection Criteria:** See **Table 8**.

Table 8: Equipment Selection Criteria

Equipment	Type and Description

Compressor Cooling Water Pumps	• Compressor Cooling Water Pumps shall be Centrifugal Type, End-Suction or Split Case Type.
	• Pump sizing and selection shall be as follows:
	  ○ Pumps shall be selected with the performance indicated at the design operating point (flow and pressure, at partial VSD speed and motor brake-horsepower).
	  ○ Pump curve shall show the performance indicated at the maximum operating point (flow and pressure, based on pump impeller size, full
	  ○ VSD speed, and full brake-horsepower).
	  ○ Pumps shall be Cast Iron casing, with Stainless Steel internally wetted parts (shaft, impeller, etc.).
	• Pump motor sizing and selection shall be as follows:
	  ○ Pump motors shall be selected for non-overloading operation on all points of the performance curve at the design conditions.
	  ○ Pump motors shall be rated for inverter duty with variable speed operation.
	  ○ Pump motor efficiency rating shall be minimum IE3.
	  ○ Pump motor IP rating shall be minimum IP55.
	• Pump installation shall be as follows:
	  ○ Each pump shall be installed with full size isolation valves, pressure gauges, flexible connections, and low point drain valves on both the return and discharge side of the pump.
	  ○ Each pump shall have a strainer or suction diffuser installed on the pump suction side, and a check valve installed on the pump discharge side.
	  ○ Each pump shall have concrete inertia base and vibration isolation springs.
	  ○ Each pump shall be installed on minimum 4-inch (100mm) high concrete housekeeping pad.

Equipment	Type and Description
Compressor Cooling Water Heat Exchangers (for heat recovery)	• Heat Exchangers shall be Plate and Frame Type. • Heat Exchanger sizing and selection shall be as follows: ○ Each Heat Exchanger shall have expansion capability to add additional plates in the future. The expansion capability shall allow for a future increase in capacity by a minimum of 20%-50% capacity. ○ Heat Exchangers shall be Carbon Steel frame with Stainless Steel plates, Stainless Steel wetted parts/piping connections, and EPDM gaskets. • Heat Exchanger installation shall be as follows: ○ Each Heat Exchanger shall be installed with temperature gauges and pressure gauges on both the supply and return side of the Heat Exchanger for hot side and cold side. ○ Each Heat Exchanger Condenser Water control valve shall have a partial-line size bypass with isolation valve, around the control valve. This is to allow maintenance and repair to be performed on the control valve, while keeping the Heat Exchanger in manual (bypass) operation. ○ Provide stainless steel containment drip pan below the heat exchangers and flange connections.

7.13 **Makeup Air Unit (MAU) and Exhaust Fan**: The equipment selection and specification requirements for Makeup Air Units and Exhaust Fans will not be covered in this System Standard.

7.13.1 The other Global System Standards for Makeup Air Units and Exhaust Fans can be referenced, for general selection and specification requirements of these units.

8.0 Piping Design and Sizing Requirements:

8.1 **Piping Layout** – CDA Distribution Main Piping shall be arranged to form a closed loop “ring main” system around the inside perimeter of the building, to improve distribution and minimize pressure drop throughout the system. CDA Distribution Laterals shall be connected to both sides of the “ring main” system.

8.2 HPCDA and XCDA typically only serve the Photolithography areas and as such, a closed loop "ring main" system is not required for HPCDA and XCDA.

8.3 **Piping Diversity and Sizing Factor** – The CDA and HPCDA main piping shall be sized according to N+F the System Diversity and Sizing factors stated in the above sections.

8.4 Each CDA and HPCDA lateral shall be sized so that the lateral's capacity meets 115-120% of the CDA and HPCDA demand at full build for the lateral (after initial construction of the system), based on the manufacturing tool layout and TUM. Therefore, each CDA and HPCDA lateral shall have 15-20% spare future capacity over the CDA and HPCDA demand at full build for the lateral (after initial construction of the system). See **Table 9**.

Table 9: CDA and HPCDA Pipe Sizing

- CDA and HPCDA Main Piping Sizing = N + F
- CDA and HPCDA Lateral Demand at Full Build, DL = TUM Peak Demand for Lateral x 70-80% Diversity
- CDA and HPCDA Lateral Piping Sizing = 115-120% x D_L

8.5 **Pipe Sizing –** CDA and HPCDA Pipe Sizing shall be selected based on the full build capacity of the CDA and HPCDA system equipment, including the future equipment (N+F). The Pipe Sizing shall be designed with maximum flowrate not to exceed the values in **Table 10**.

Table 10: Pipe Sizing Criteria

CDA / HPCDA / XCDA/ IA Nominal Pipe Size	Maximum Flowrate			Approximate Pressure Loss
	SCFM	NCMH	SLPM	
2" DN50	270	459	7,650	0.50 psi/100 ft
2.5" DN65	594	1,010	16,820	0.50 psi/100 ft
3" DN80	810	1,377	22,950	0.50 psi/100 ft
4" DN100	1,420	2,414	40,200	0.33 psi/100 ft
6" DN150	3,600	6,120	101,950	0.25 psi/100 ft
8" DN200	8,000	13,600	226,550	0.25 psi/100 ft
10" DN250	14,500	24,650	410,600	0.25 psi/100 ft
12" DN300	23,500	39,950	665,450	0.25 psi/100 ft

CDA / HPCDA / XCDA/ IA Nominal Pipe Size	Maximum Flowrate			Approximate Pressure Loss
	SCFM	NCMH	SLPM	
14” DN350	36,000	61,200	1,019,400	0.25 psi/100 ft
16” DN400	51,500	87,550	1,458,300	0.25 psi/100 ft
18” DN450	71,000	120,700	2,010,500	0.25 psi/100 ft
20” DN500	94,500	160,650	2,675,950	0.25 psi/100 ft
22” DN550	122,000	207,400	3,454,650	0.25 psi/100 ft
24” DN600	154,500	262,650	4,374,950	0.25 psi/100 ft
26” DN650	191,500	325,550	5,422,700	0.25 psi/100 ft

8.6 Materials of Construction

8.6.1 CDA and HPCDA Distribution piping shall be 304L Schedule 10 Stainless Steel piping or tubing, and shall meet the following spec:
- Free of oil and grease per CGA G-4.1-2004 Standard
- Mechanically polished to achieve Standard Roughness Average (RA) of 25 to 40 μ-in for internal surface of piping
- **Note:** Cleaned for Oxygen Service (CFOS) material spec is not required for CDA, HPCDA, XCDA, and IA piping.

8.6.2 CDA and HPCDA compressor vent/blow-off piping and relief valve vent piping shall be ASTM A53 Schedule 40 Carbon Steel piping.

8.6.3 Condenser Water piping for Compressor Cooling Water shall be ASTM A53 Schedule 40 Carbon Steel piping or ASTM A53B Schedule 40 Galvanized Steel piping.

8.6.4 XCDA Distribution piping shall be 316L Schedule 10 Stainless Steel tubing, and shall meet the following spec:
- Free of oil and grease per CGA G-4.1-2004 Standard
- Electropolished to achieve Standard Roughness Average (RA) of 10 to 20 μ-in for internal surface of piping
- Bright Annealed (BA)

8.6.5 IA Distribution piping shall be 304L Schedule 10 Stainless Steel or Type L Hard Copper Piping, and shall meet the following spec:
- Free of oil and grease per CGA G-4.1-2004 Standard
- Mechanically polished to achieve Standard Roughness Average (RA) of 25 to 40 μ-in for internal surface of piping

8.6.6 Piping sections shall be joined via welded ends as much as possible, to minimize the number of flanges and mechanical joints in the system.

8.6.7 Mechanical joints shall be flanged connections.

- **Note**: Threaded connections, unions, push-fittings, crimped-fittings, and quick-connect fittings shall not be used on Compressed Air systems.

8.7 **Spare Valves for Main Expansion** – Full size isolation valves shall be provided at the ends of all main piping headers for equipment (i.e. main piping headers for Compressor CDA Discharge, Receiver Tank Inlet and Discharge, Dryer/Filter Inlet and Discharge, and CCW Supply and Return). Also, a minimum of three (3) Full size isolation valves shall be provided at the main CDA and HPCDA piping supply headers, immediately after the Final Filter Discharge. The future valves are to allow future capacity expansion or cross-connections, etc.

8.8 **Spare Valves for Temporary Rental Compressor Equipment** – Full size isolation valves shall be provided on the main CDA and HPCDA piping headers, both upstream of the Dryer/Filters and downstream of the Dryer/Filters, with close proximity to a nearby perimeter wall, to allow for future connection to temporary Rental Compressor Equipment outside.

8.9 **Spare Valves for Future Laterals** - Additional isolation valves shall be included on the main supply and return piping distribution piping, to allow installation of future laterals for capacity expansion. Based on the number of laterals included in the design, a minimum quantity of 35-50% of additional isolation valves for future laterals shall be included.

8.10 **Lateral Point-of-Connection Quantity and Sizes** – The laterals shall have sufficient quantity of Point of Connections (POCs) to meet the tool layout and requirements.

8.10.1 A minimum quantity of 30-40% of additional spare POCs shall be provided on each lateral, for future tools.

8.10.2 The sizes for each POC shall be checked and verified versus the actual tool requirements. Minimum POC size shall be DN25 (1”), but larger POC sizes can be provided, based on the actual tool requirements.

8.11 **Sampling Ports** – Sampling ports shall be installed on the ends of all main piping sections and all laterals, to allow air sampling and quality checks.

8.12 **Receiver Tank Bypass** – There shall be full line-size bypass with isolation valve around the CDA and HPCDA receiver tanks, to allow isolation of the tanks during inspection and maintenance.

9.0 HVAC for Compressor Air Intake and Exhaust

9.1 **MAU Units** – 100% Outside Air Makeup unit(s) shall provide conditioned and filtered supply air for the Compressor Air Intakes. There shall be N+1 redundancy for the MAU Units.

9.1.1 The MAU units provide supply air to the Compressor Air Intakes at lower supply temperature and lower moisture content, for more efficient Compressor and Dryer operation. Also, the supply air is filtered, to reduce the loading on Compressor Air Intake filters and improve CDA air quality.

9.1.2 The location of the MAU fresh air intake shall be located a minimum of 5m (16.4 ft) away from any exhaust discharge or contaminants source.

9.2 **MAU Supply Ductwork Terminations at Compressor Air Intake** – Terminate supply air ductwork with open-ended hoods, at approximate height of 0.5m above each Compressor air intake connection.

9.2.1 The open-ended hoods ensure that if the MAU units trip offline or are not operating, the Compressors will still have sufficient intake air to prevent the Compressors from surging or tripping offline.

9.3 **Exhaust Fans** – Exhaust fan(s) shall remove heat from the equipment (Compressors, Dryers, etc.), and discharge to the outside. There shall be N+1 redundancy for the Exhaust fans.

9.4 **Exhaust Ductwork Terminations at Compressor Heat Exhaust** – Terminate exhaust ductwork with open-ended hoods, at approximate height of 0.5m above each Compressor heat exhaust connection. Branch exhaust ductwork for Dryers can be connected directly to the dryers.

9.4.1 The open-ended hoods ensure that if the exhaust fans trip offline or are not operating, the Compressor heat exhaust can still be discharged from the Compressors.

9.5 **Exhaust Ductwork Terminations at Air Dryer Regeneration and Heat Exhaust** – Connect branch exhaust ductwork directly to the Air Dryers, with flexible connections. The flexible connection material shall be rated for the air temperature that is discharged from the Air Dryers.

10.0 ICA Supply Manifold and N2 Backup:

10.1 **ICA Supply** – A dedicated ICA main pipe shall be branched from the CDA main supply header downstream of the CDA dryers and filters, to provide compressed air for the ICA system. The ICA main pipe and system shall include the following:

10.1.1 Main isolation valves
10.1.2 Dual check valves
10.1.3 There shall be parallel manifold with dual pressure regulator stations, each with isolation valves, and a full-size bypass line with isolation valve. The pressure regulators shall reduce the ICA supply pressure to 0.65 MPa (94.3 psig) at the main supply.
10.1.4 Automatic control valve with isolation valves and bypass line, to use for isolation of CDA upon N2 backup supply to ICA.
10.1.5 ICA distribution piping with laterals and POCs

10.2 **N2 Backup to ICA –** A full size N2 Backup line shall be connected from Bulk Gas N2 supply to the ICA main pipe. The N2 Backup shall be used to maintain an uninterrupted supply of ICA, in case of any full system loss or significant pressure loss from the Primary CDA system serving ICA. The N2 Backup line shall include the following:

10.2.1 Main isolation valves
10.2.2 Dual check valves
10.2.3 Automatic control valve with isolation valves and bypass line, to open and supply N2 as backup to ICA, upon a full system loss or significant pressure loss from CDA/ICA
10.2.4 Automatic control valve with isolation valves and bypass line, to use for isolation of CDA upon N2 backup supply to ICA (see Section 10.1.5)

11.0 Accessories:

11.1 The following accessories which shall be installed: See **Table 11.**

Table 11: Accessories to be installed

Accessories	Type and Installation Location	Remarks
Piping Insulation	• Install for all CHW piping throughout the system	In accordance with Spec Section 15260 (or equivalent Spec)
Ductwork Insulation	• Install for all MAU Supply Air ductwork • Install for CDA compressor heat exhaust or CDA Air Dryer heat exhaust that exceeds 40°F (104°C) surface temperature.	
Piping Supports	• Install piping hangers, supports, racks, and clamps at the appropriate spacing interval as required by the system and piping load and layout • Provide seismic restraints and supports as deemed required during design, based on site location and local jurisdiction requirements • Provide vibration isolation for piping supports as deemed required during design, based on the building vibration criteria	In accordance with Spec Section 15140 (or equivalent Spec)
Flexible Connections	• Install at all CCW pump connections on suction and discharge • Install at all CHW connections to Heat Exchanger on supply and return	

Accessories	Type and Installation Location	Remarks
Pressure Gauges	• Install at each Compressor discharge • Install at each Dryer discharge • Install at each Receiver Tank • Install Differential Pressure gauge on each filter • Install at each supply lateral end-of-line for CDA, • HPCDA, XCDA, ICA • Install at ICA main pipe connection from CDA Install at each Pressure Regulating Valve upstream and downstream • Install at Aftercooler supply and return connection for CHW • Install at CCW pump suction and discharge • Install at Compressor supply and return connection for CCW	
Temperature Gauges	• Install at each Compressor discharge for CDA Install at each Aftercooler inlet and outlet connection for CDA and HPCDA • Install at each Aftercooler supply and return connection for CHW • Install at each Compressor supply and return connection for CCW	
Air Vents	• Air vents shall be minimum size DN20 (3/4 inches) • Install Auto or Manual Air Vents at all high point locations for CHW and CCW • Install Auto Air Vents at main CHW piping and vertical CHW piping connected to the Aftercooler • Install Auto Air Vents at main CCW piping and vertical CCW piping connected to CCW Pumps	
Drain Ports	• Install at all low point locations • Install at all Compressors • Install at all Aftercoolers • Install at all Receiver Tanks • Install at all Dryer/Filters • Install at all CCW Pumps	

Accessories	Type and Installation Location	Remarks
Valves	• All butterfly valves shall be rated for "bi-directional" duty with positive shutoff in both directions • All butterfly valves shall be lug-style	In accordance with Spec Section 15101 (or equivalent Spec)
Control Valves	• Provide control valve type and quantity as outlined in this standard	
Relief Valves	• CDA Receiver Tank shall have dual pressure relief valves rated at 1.05 MPa (152.3 psig) each. • HPCDA Receiver Tank shall have dual pressure relief valves rated at 1.15 MPa (166.8 psig) • Main ICA piping shall have dual pressure relief valves rated at 0.86 MPa (124.7 psig) each.	
Condensate Drain	• Drain piping shall be provided to collect condensate from the CDA system, and to discharge to the Industrial Waste (IW) system or other approved system.	

Accessories	Type and Installation Location	Remarks
Labeling and Identification	• "CDA" nomenclature shall be used for Clean Dry Air • "HPCDA" nomenclature shall be used for High Pressure Clean Dry Air (and Photo Clean Dry Air) • "XCDA" nomenclature shall be used for Purified Clean Dry Air • "ICA" nomenclature shall be used for Instrument Control Air • "CCWS" and "CCWR" shall be used for Compressor Cooling Water Supply and Return • "CHWS" and "CHWR" shall be used for Chilled Water Supply and Return • Provide identification labels at minimum of every 6m distance and change of direction • Labels for CDA, HPCDA, XCDA, and ICA piping shall be Blue Background with White Lettering • Labels for CCWS, CCWR, CHWS, and CHWR piping shall be Green Background with White Lettering	In accordance with Spec Section 15190 (or equivalent Spec)

12.0 Electrical Design:

12.1 Electrical power supply system for Compressors, Dryers, Cooling Water Pumps, and all other equipment shall be designed and installed in compliance with **Global Facilities - Design & Construction Standard - Electrical - Electrical Design Specification**.

12.2 **Power Segregation** – Electrical power supply for Compressors, Dryers, Cooling Water Pumps, and all other equipment shall be segregated among a minimum of 2 separate upstream power sources (including panel + transformer). The quantity of equipment served from each upstream power source shall be split evenly.

12.3 **Emergency Power for Compressor** – At least 1 no. Compressor (CDA & HPCDA) shall be connected to an Emergency Power source (such as Electrical Generator, etc) to maintain pressurization of distribution piping to prevent potential contamination during power outage.

12.4 **Emergency Power for Air Dryers** – All air dryers (CDA & HPCDA) shall be connected to an Emergency Power source (such as Electrical Generator, etc) to maintain pressurization of dryer desiccant bed to prevent potential contamination during power outage, and to avoid lengthy recovery of dryer air condition back to meet dew point specification.

12.5 **Emergency Power for Compressor Cooling Water Pumps** – At least 2 no. Cooling Water Pumps (CDA & HPCDA) shall be connected to an Emergency Power source (such as Electrical Generator, etc) to maintain cooling water circulation to Compressor during power outage.

12.6 **Un-interrupted Power (with Emergency Power Back-up) for Controls** – 100% of CDA system controls including PLCs, Local Control Panels (LCPs), CDA Central Controller, Compressor Control Power, Dryer Control Power, and other auxiliary control power shall be connected to an Uninterrupted Power source (such as UPS or DC Bank) to allow continuous operation of the CDA system during momentary power interruption or voltage sags, spikes, and fluctuation events.

12.7 **Variable Frequency Drives** – CCW pumps shall be VFD driven for variable speed operation. Variable Frequency Drives shall be selected to be one size larger than the power rating of the actual CCW pump motor size.

12.4.1 Variable Frequency Drives shall be compliant with SEMI standard F47 for voltage sag immunity, to provide ride-through capability during momentary power interruption or voltage sag.

12.8 **Local Disconnect Switches** – Each CCW pump shall have a lockable type, local disconnect switch installed adjacent to the pump, to allow for manual isolation and lockout/tagout (LOTO/COHE) of the pump.

13.0 Instrumentation:

13.1 Instrumentation for Compressors, Dryers, Cooling Water Pumps, and all other equipment shall be designed and installed in compliance with **Global Facilities - Design & Construction Standard - I&C – Instrumentation**.

13.2 The following Instrumentation shall be installed and monitored/ controlled via FMCS. See **Table 12.**

Table 12: Instrumentation to be installed

Instrumentation	Type and Installation Location
CDA and HPCDA Pressure Transmitters	• Install at Air Receiver Tank(s) (Quantity: 1 duty and 1 backup [2 Total]) • Install at Main Supply Header after Final Filters (Quantity: 1 duty and 1 backup [2 Total]) • Install at Supply Lateral end-of-line, at minimum 2 locations (Quantity: Minimum 2 locations) • Install at locations of segregated buildings or critical loads
CDA and HPCDA Dewpoint Temperature Transmitters	• Install at Main Supply Header after Final Filters (Quantity: 1 duty)
CDA and HPCDA Dry Bulb Temperature Transmitter	• Install at Main Supply Header after Aftercoolers (Quantity: 1 duty and 1 backup [2 Total]) • Install at each outlet of Dryers (Quantity: 1 duty for each dryer)
Compressor Cooling Water Heat Exchanger Temperature Transmitter	• Install at Main Compressor Cooling Water Supply Header after CCW Heat Exchangers (Quantity: 1 duty and 1 backup [2 Total])
CDA Supply Flowmeters	• Install at Main CDA Supply Header after Final Filters (Quantity: 1 duty)
HPCDA Supply Flowmeters	• Install at Main HPCDA Supply Header after Final Filters (Quantity: 1 duty)
ICA Pressure Transmitter	• Install at Main ICA Supply Header after main takeoff from CDA (Quantity: 1 duty)
ICA Supply Flowmeter	• Install at Main ICA Supply Header after main takeoff from CDA (Quantity: 1 duty)
Additional Spare Ports	• All supply and return laterals shall have a spare port and spare valve installed at the lateral end-of-line, to allow for future pressure transmitters to be installed. The future pressure transmitters will allow continuous SCADA monitoring of each lateral pressure.

14.0 Control Sequence Philosophy:

14.1 The overall philosophy for system control and operation shall be listed in this section. See **Table 13**. Refer to the specific Sequence of Operations for each project for specific details.

Table 13: System Control Logic

Control Parameter	Control Logic
CDA Pressure Control	• CDA Compressor Discharge pressure shall be controlled by via the Compressor Central Controller. ○ Controller shall determine the best operating combination and setpoints of the Compressors, to maximize energy efficiency and regulate stable system pressure within a narrow pressure band ○ Controller shall determine Compressor workload based on run-time equalization ○ Controller shall auto-start next Compressor if a running Compressor faults or trips offline ○ The Central CDA system pressure shall be measured at the pressure transmitters mounted on the Air Receiver Tank. • As a backup means, CDA Compressor Discharge pressure can also be controlled based on the standalone pressure control of each Compressor, from the Local Control Panel located at each Compressor
Aftercooler Temperature Control	• Position output of Aftercooler chilled water control valves shall be modulated to maintain the CDA aftercooler leaving air temperature setpoint (as measured by main temperature transmitters, located downstream of Aftercoolers)
Compressor Cooling Water Heat Exchanger Temperature Control	• Position output of CCW Heat Exchanger condenser water control valves shall be modulated to maintain the CCW Heat Exchanger leaving water temperature setpoint (as measured by CCW Heat Exchanger temperature transmitters, located downstream of Heat Exchanger)
CDA Dewpoint Control	• CDA Supply Dew Point temperature shall be maintained at each Dryer Local Control Panel. The Dryer shall adjust its operation (regeneration, purge) function and switch online/offline towers based on Dewpoint demand control. The main CDA Supply Dew Point transmitters, as measured by the main dewpoint transmitters located on the main supply header, shall be used for SCADA monitoring and alarm only.

Control Parameter	Control Logic
ICA Pressure Control via N2 Backup	• When the ICA pressure (as measured by the main ICA pressure transmitters) drops below the minimum pressure setpoint, the N2 Backup to ICA Control Valve shall open to maintain the ICA pressure and keep uninterrupted supply of ICA. The ICA Main CDA Isolation Control Valve shall close, after it is confirmed that the N2 Backup to ICA Control Valves have opened.

Note: The control philosophy for the MAUs, Exhaust Fans, and CCW Pumps will not be covered in this Standard.

14.2 The following table shows the control valve fail positions that shall be configured. See **Table 14.**

Table 14: Control Valve Fail Positions

Control Valve	Control Valve Fail Position		
	Loss of Air	Loss of Electricity	Loss of Control Signal
Aftercooler Temperature Control Valve	Fail Last State	Fail Last State	Fail Last State
Compressor Cooling Water Heat Exchanger Temperature Control Valve	Fail Last State	Fail Last State	Fail Last State
ICA Main CDA Isolation Control Valve	Fail Last State	Fail Last State	Fail Last State
N2 Backup to ICA Control Valve	Fail Last State	Fail Last State	Fail Last State

14.3 Any control valve actuators configured to fail-last state shall have the capability of failing to last state position and holding the position for unlimited duration without needing control air, power, or PLC signal.

14.4 Upon a loss event of control air, power, or PLC signal, and then after the utility is restored, the control valve output command/position shall be set to the same command/position it was operating at prior to the loss event.

14.5 All control valves shall have a manual override bypass, to allow manual operation of the valve upon a failure.

15.0 Smart Facilities Design:

15.1 The following are additional enhancements that can be considered and included to enable Smart Facilities Design concepts.

15.1.1 **Lateral Pressure Online Monitoring –** Consider installing pressure transmitters at all CDA, HPCDA, XCDA, and IA laterals at each end-of-line location, to allow continuous SCADA monitoring and alarm of each lateral pressure.

15.1.2 **Load Demand Online Monitoring** – Consider installing flowmeters at each building/area or each major load/user, to allow continuous SCADA monitoring and tracking of CDA, HPCDA, and XCDA load and demand each different user.

15.1.3 **Compressor Vibration Online Monitoring** – Consider installing vibration sensors on each Compressor and motor, to allow continuous monitoring and alarm of Compressor vibration levels. Some Compressors already have Factory Mounted, vibration sensors and monitoring software integral to the equipment.

15.1.4 **Compressor Cooling Water Pump Vibration Online Monitoring** – Consider installing vibration sensors on each Pump and motor, to allow continuous monitoring and alarm of Compressor vibration levels.

16.0 Inter-Discipline Coordination Matrix:

16.1 The following table shows the provisions that shall be provided by each of the following disciplines and coordinated accordingly. See **Table 15.**

Table 15: Provisions Provided by Each Discipline

Discipline	Provision
Architectural	• Provide dedicated Utility Room or Mechanical Room space for centralized CDA equipment • Provide concrete housekeeping pad below all equipment including Compressors, Air Receiver Tanks, Aftercoolers, Filters, Dryers, etc. • Cut openings in walls and floors for piping penetrations, and seal openings after piping installation • Provide adequate fire stopping at all required fire wall penetrations
Mechanical	• Provide HVAC space conditioning for dedicated Utility Room or Mechanical Room space containing the centralized CDA equipment. Room temperature shall be designed to maintain the space temperature range of 24°C to 28°C. • Provide Makeup Air Unit(s), heating water and chilled water supply piping, supply ductwork, and accessories as outlined in this Standard, to provide sufficient air supply for the CDA Compressor Intake air • Provide Exhaust Fan(s), exhaust ductwork, and accessories as outlined in this Standard, to provide sufficient exhaust for the CDA Compressor Heat Exhaust and Dryer purge exhaust. • Provide chilled water piping and accessories as outlined in this Standard • Provide condenser water piping, and accessories as outlined in this Standard • Provide floor drains located suitably to allow proper draining of equipment, condensate drains and auto-drains, piping, and air vents

Discipline	Provision
Electrical	• Provide VFDs for all CCW pumps • Provide local disconnect switches for all CCW pumps, Exhaust Fans, MAU • Provide power wiring for all Electrical equipment/components • Provide grounding for all Electrical equipment/components • Provide Emergency Power / UPS Power as outlined in this Standard • Provide Power Segregation as outlined in this Standard • Provide Voltage transformer(s), if required for CDA VSD Screw Compressors
Water / Wastewater	• Provide drain source for the Condensate Drain water piping
Controls	• Provide control valve actuators and positioners • Provide control wiring • Provide instrument control air tubing • Provide FMCS/SCADA system monitoring and controls of all instrumentation and transmitters including pressure, temperature, dewpoint, flow, etc. • Integrate CDA and HPCDA Central Controller with FMCS/SCADA

17.0 Quality Inspection and Testing:

17.1 **Inspection and Test Plan (ITP) –** Create formal Inspection and Test Plans (ITP) that outline all requirements for quality verification, inspections, testing, and commissioning:

17.1.1 Piping and Ductwork

- Piping and Ductwork Shop Drawings and Submittals
- Incoming Material Inspection
- Material Storage
- Welding Procedures, Qualifications, and Inspection
- Piping, Ductwork, and Supports Installation and Verification
- QA/QC Inspection & Punchlist
- Leak Testing and Pressure Testing
- Cleaning, Flushing, and Purging
- Quality Certification
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

17.1.2 Equipment

- Equipment Shop Drawings and Submittals
- Factory Acceptance Test
- Incoming Equipment Inspection
- Equipment Move-In and Installation
- QA/QC Inspection & Punchlist
- Equipment Pre-Startup Checks – Mechanical, Electrical, Controls, etc.
- Operational Acceptance Test (OAT) - Standalone Equipment Startup per Vendor's Procedure to allow equipment to run in MANUAL mode. Also considered "Early Run" Equipment.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

17.1.3 System

- Site Acceptance Test (SAT) – Test System operation and control functions in accordance with P&ID, SOO, and Specifications.
- Integrated System Test (IST) – Test full System operation including all system interdependencies, interlocks, and critical failure modes.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

17.2 **QA/QC Inspection** – Prior to leak testing and pressure testing, QA/QC inspection shall be performed on all installed piping and components to verify proper installation in accordance with construction drawings, project specifications, and industry standards.

17.3 **Gross Leak Test** - Gross leak test shall be performed via compressed Nitrogen or Air at pressure up to 5 psig. Walk the entire system to verify that no leaks are present, and to make repairs to any leaks that are identified.

17.4 **Pressure Test** - All main piping shall be pneumatic pressure tested at a minimum of 1.1-1.3 times the design operating pressure, for a minimum of 24 hours, with zero pressure loss. Submains and laterals shall be pressure tested for a minimum of 8 hours, with zero pressure loss.

17.4.1 **CDA:** Pneumatic pressure test to be performed at minimum 1.24 MPa (180.0 psig)
17.4.2 **HPCDA:** Pneumatic pressure test to be performed at minimum 1.35 MPa (195.0 psig)
17.4.3 **XCDA:** Pneumatic pressure test to be performed at minimum 1.24 MPa (180.0 psig)
17.4.4 **ICA:** Before Pressure Regulators: Pneumatic pressure test to be performed at minimum 1.24 MPa (180.0 psig)

Note: For ICA After Pressure Regulators: Pneumatic pressure test to be performed at minimum 0.85 MPa (123.5 psig)

17.5 **Flushing and Purging** – After completion of installation and pressure test, the piping shall be purged via compressed Nitrogen or Air at a velocity of 50 FPS to clean and flush the piping, until the Quality Certification requirements can be met.

17.6 **Quality Certification Test** – Particle testing, moisture content testing, and oil content testing shall be performed on all piping, to meet the Quality Certification requirements outlined in this standard.

18.0 Startup Commissioning / Site Acceptance Test

18.1 At a minimum, the following system control functions shall be tested and successfully proven during the System Startup Commissioning / Site Acceptance Test.

18.1.1 Compressors

- Verify Standalone Commissioning for each Compressor, per manufacturer's instructions
- Verify Compressor Start/Stop Function to Rotate Operating Compressor Units
- Verify Compressor Auto-Start sequencing, upon failure of an operating Compressor
- Confirm system pressure control is stable during Compressor manual start/stop rotation
- Confirm system pressure control is stable during Compressor auto-start sequencing, upon failure of another Compressor
- Confirm Compressor Rotation Function, Auto-Sequencing, and stable system pressure control via the Master Sequencer Central Control panel

18.1.2 Dryers

- Verify Standalone Commissioning for each Dryer, per manufacturer's instructions

18.1.3 Aftercoolers

- Verify Aftercooler CHW Control Valve Function through the operating range (0 to 100%)
- Confirm Aftercooler CHW Control Valve control in response to compressed air supply temperature fluctuations

18.1.4 CCW Pumps

- Verify Standalone Commissioning for each CCW Pump, per manufacturer's instructions
- Verify CCW Pump Start/Stop Function to Rotate Operating CCW Pumps
- Verify CCW Pump Auto-Start sequencing, upon failure of an operating Pump
- Verify CCW Pump Speed control in response to CCW pressure fluctuations

18.1.5 CCW Heat Exchanger

- Verify Standalone Commissioning for each CCW Heat

Exchanger, per manufacturer's instructions

- Verify CCW Heat Exchanger CHW Control Valve Function through the operating range (0 to 100%)
- Confirm CCW Heat Exchanger CHW Control Valve control in response to CCW supply temperature fluctuations

18.1.6 N2 Backup to ICA

- Confirm N2 backup to ICA is functional upon a drop in CDA/ICA supply pressure, and that N2 backup can maintain the required pressure
- Confirm N2 backup to ICA is functional upon a total loss of CDA supply, and that N2 backup can maintain the required pressure

18.1.7 Instrumentation and Controls

- Confirm Auto-switch to backup CDA/HPCDA pressure transmitter upon failure of main pressure transmitter
- Confirm Auto-switch to backup CDA/HPCDA temperature transmitter upon failure of main temperature transmitter
- Confirm Auto-switch to backup CCW pressure transmitter upon failure of main pressure transmitter
- Confirm Auto-switch to backup CDA/HPCDA Aftercooler temperature transmitter upon failure of main temperature transmitter
- Confirm Auto-switch to backup CCW Heat Exchanger temperature transmitter upon failure of main temperature transmitter
- Confirm all control valves fail to the stated position per this Standard, upon loss of instrument air, power, and/or control signal.
- Verify all alarms are enabled and activated in SCADA

19.0 Reference Specifications:

19.1 The following Construction Specifications shall be referenced during design, engineering, and construction. The Specifications shown below are in accordance with the Construction Specifications Institute (CSI) Master Format; Refer to the specific Construction Specifications for each project. See **Table 16**.

Table 16: Reference Construction Specifications

Spec Section	Spec Contents	Remarks
Section 15485	Clean Dry Air	
Section 15140	Piping Supports and Anchors	
Section 15190	Piping Labels and Identification	
Section 15260	Piping Insulation	
Section 15280	Equipment Insulation	
Section 15515	Hydronic Specialties	
Section 15990	Testing, Adjusting and Balancing	
Section 15170	Motors	
Section 15172	Variable Frequency Drives	
Section 15070	Stainless Steel Piping	
Section 15101	Valves	
Section 15481	Air Dryers	
Section 15485	Air Compressors	
Section 15540	Pumps	
Section 11223	Filters	
Section 15242	Vibration Isolation	
Section 15695	Steel Tanks	

20.0 Document Control

20.1 **Approval**: GLOBAL_FAC_GFTT_APPR

20.2 **Notification**:
GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_ NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIF

20.2 **Final Viewer Access**
All Micron sites Facilities Team Members

20.3 **Retention**
Adherence to the requirements specified in the
Global Facilities - Document Control Procedures

20.4 **Review**
Biennially (two years)

21.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	03/27/2020
1		Revised Document	03/21/2022
2	Section 7	7.10 Updated sub-section 7.11.2 Added new sub section 7.11.3 Added new sub section 7.11.4 Added new sub section	11/01/2022
	Section 12	12.3 Added new sub-section 12.4 Added new sub-section 12.5 Updated sub-section 12.6 Added sub-section	
	Appendix	Updated Appendix 2	

Appendix 1: Template Form for Site Specific Deviation List

Site:______________________

No	Specification Section and Item	Deviation Proposal	Scope Impact (Benefit, Cost, Schedule)	Remark	Approval: 1. Site 2. GFTT

Appendix 2: Recommended provision of Compressed Air for pneumatic equipment of various Utility Systems

No	System	Area / Equipment	Compressed Air Provision
1	Specialty Gases	Gas Room: GC, Abatement, etc (AOV)	GN2
		Sub Fab: GC, VMB, DVB, Actuated Valves (AOV)	GN2
2	Chemical/ Slurry	Chemical Room: Chemical System Diaphragm Pump	ICA
		Chemical Room: Chemical System AOV	ICA
		Sub-Fab: Chemical VMB POU AOV	GN2
		Chemical Room: Chemical System Levitronix Cooling Air	ICA
		Slurry room / Sub-fab: Slurry System Diaphragm Pump	ICA
		Slurry Room / Sub-fab: Slurry System AOV	ICA
		Sub-fab: Slurry VMB POU AOV	GN2
		Sub-fab: Slurry System Levitronix Cooling Air	ICA
3	Mechanical	CHW System Control Valves	ICA
		PCW System Temperature Control Valves	ICA
		Hot Water Control Valves	ICA
		Process Exhaust Systems Dampers & Control Valves	ICA
		MAU/AHU Temperature Control Valves / Dampers	ICA
		Cleanroom DCC Temperature Control Valves	ICA
4	Water	UPW System	ICA
		WWTP System	ICA
		Lift Station System	ICA
		Waste Chemical Collection System	ICA
		i. Sub fab	GN2
		ii. Waste chemical room	ICA
		iii. Chemical room	ICA

** Note: ICA shall be used as the compressed air source for the facilities utility supply by referring to **Appendix 2**. Thus, 100% backup of CDA supply via N2 is not required and shall not be provided.